Medical Triage Knowledge Graph

A Multi-Ontology GraphRAG Pipeline for Emergency Department Decision Support

LMP2392 — Healthcare and Medicine for AI Experts

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Outline

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The Clinical Problem

Emergency Department triage is time-critical

- Clinicians must rapidly map presenting symptoms to the right diagnostic test under cognitive and time pressure.
- Clinical guidelines (AHA/ACC, ACR, NICE, CTAS) encode evidence-based rules — but they are dispersed across long PDFs and multiple organisations.
- LLMs alone can *hallucinate* clinical pathways; they lack grounded, citable reasoning.

Our answer: a knowledge graph as the ground truth layer.

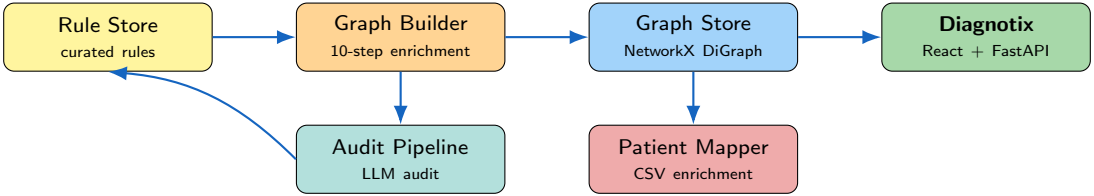
Target pathways

- ECG
- Testicular Ultrasound
- Arm X-Ray
- Abdominal Ultrasound
- Appendix Area Ultrasound

Guideline sources

AHA/ACC ACR NICE CTAS

High-Level Architecture



Rules are iteratively audited and revised until satisfactory, then the graph builder serialises the final graph. The graph store powers **Diagnotix** for dynamic expansion and the patient mapper for CSV enrichment.

Source of Truth: The Clinical Rule Store

Each of the 45 rules encodes a single clinical assertion:

$$\texttt{raw_symptom} \longrightarrow \texttt{condition} \longrightarrow \texttt{test}$$

Example rule (JSON)

```
{
  "raw_symptom": "chest pain",
  "node_type": "Symptom",
  "condition": "Acute MI",
  "test": "ECG",
  "source": "AHA/ACC"
}
```

Rule fields

Field	Meaning
raw_symptom	Clinical entity
node_type	KG category
condition	Target diagnosis
test	Recommended test
source	Guideline body

10-Step Enrichment Pipeline (Graph Builder)

Step	API / Source	Enriches	Column(s) added
0	Static rules	—	base DataFrame
1	URL Registry	`(source, test)`	`url`
2	EMBL-EBI OLS4 (HP/MONDO/EFO)	`raw_symptom`	`ebi_code`, `synonyms`
3	Infoway FHIR (SNOMED CT CA)	`raw_symptom`	`infoway_code`
4	UMLS REST (LOINC)	`test`	`loinc_code`
5	UMLS REST (ICD-10 international)	`condition`	`icd10_code`
6	Infoway FHIR (ICD-10-CA)	`condition`	`icd10ca_code`
7	NLM RxNorm + OpenFDA	`treatment`	`rxcui`, `adverse_event`
8	Europe PMC	symptom $\leftrightarrow$ condition	`literature_weight`
9	Europe PMC	condition $\leftrightarrow$ test	`test_literature_weight`
10	ClinicalTrials.gov v2	condition $\leftrightarrow$ treatment	`trial_count`

Steps 8–10 attach **real-world evidence counts** that drive edge-width scaling in the HTML visualisation.

Graph Schema

Primary node types

Symptom	`Symptom:`
Condition	`Condition:`
Test	`Test:`

Secondary node types

Vital	`Vital:`
Demographic	`Demographic:`
Risk Factor	`Risk Factor:`
MOI	`MOI:`
Attribute	`Attribute:`
Treatment	`Treatment:`

Edge types

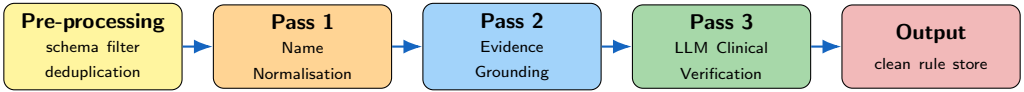
`INDICATES_CONDITION`	entity → condition
`REQUIRES_TEST`	condition → test
`DIRECTLY_INDICATES_TEST`	entity → test
`RECOMMENDS_TREATMENT`	condition → drug

Edge metadata on every edge:

- `source` — guideline body (AHA/ACC, ACR, ...)
- `source_url` — citable DOI or page URL
- evidence weight (literature / trial count)

Audit Pipeline Overview

Problem: LLM-generated rules can be fabricated, vague, or misattributed. The audit pipeline cleans the rule store before the graph is built.



Pre-processing
Drop rules with missing fields or unknown sources. Remove exact duplicate triples from repeated generation passes.

Pass 1 — Normalise
Replace LLM-invented names with canonical labels *before* LLM review: ICD-10 international (UMLS) and HP/MONDO (OLS4).

Pass 2 — Ground
Fetch Europe PMC article counts + DuckDuckGo snippets. Tag each rule with a PMC evidence tier.

Pass 3 — Verify
LLM reviews each pathway in batches of ≤ 15. Evidence tier sets the retention bias.

Audit Pipeline — Evidence Grounding & Verification

Pass 2 — Evidence Grounding

Two signals fetched per rule:

- **Europe PMC** co-occurrence count
- **DuckDuckGo** web snippets

Rules tagged by **PMC evidence tier**:

STRONG	$\geq$5 000 articles
GOOD	$\geq$500
MODERATE	$\geq$50
LIMITED	$>$0
NONE	0

Zero-evidence rules flagged for clinician review.

Pass 3 — LLM Clinical Verification

LLM reviewer grouped by pathway, batched at $\leq$**15 rules/call**.

Evidence tier drives retention bias:

- STRONG / GOOD $\rightarrow$ retain by default
- MODERATE / LIMITED $\rightarrow$ normal scrutiny
- NONE $\rightarrow$ bias to remove

Three verification criteria:

1. Source accuracy
2. Clinical plausibility
3. Condition specificity

Data Processing Pipeline — Two-Phase Overview

Goal: ingest a real patient CSV and enrich every diagnosis with KG-derived diagnostic tests.

Phase 1 — Extract candidate tests

Test Extractor

Reads all unique diagnoses from the `dx` column and asks GPT-5 Nano to identify the standard diagnostic test(s) for each. Writes a diagnosis-to-test mapping file.

Phase 2 — Expand graph & map patients

Enrichment & Mapping Pipeline

Adds any new test pathways to the KG, rebuilds the graph with full ontology enrichment, then maps every diagnosis back to a condition node and its recommended tests.

Standalone mapping (graph already current)

Patient Mapper — skips graph expansion; semantically maps a new patient CSV against the existing graph store via GPT-5 Nano.

Diagnotix — Dynamic Knowledge Graph Builder

Full-stack web application (FastAPI + React)

- Enter any diagnostic test name; the LLM extracts 8–15 evidence-based triage rules automatically.
- Generated rules are validated via an evidence-grounded audit pipeline: Europe PMC co-occurrence + DuckDuckGo web search ground each rule before LLM clinical review.
- Validated rules are appended to the rule store and the full ontology enrichment pipeline re-runs.
- Every expansion is auditable: the response reports new rule count and updated node/edge totals.

Stack

Backend: FastAPI + Uvicorn
Frontend: React 19 + TypeScript + Vite
LLM: GPT-5 Nano

Node tooltip codes

HP/MONDO	SNOMED-CT
ICD-10* (linked)	ICD-10-CA
LOINC	RxCUI

*links to WHO ICD-10 browser

Summary

What we built

- An expandable knowledge graph spanning 5 clinical pathways, grounded in evidence-based guidelines.
- 10-step ontology enrichment pipeline (SNOMED CT CA, LOINC, ICD-10 international, ICD-10-CA, HP/MONDO, RxNorm, Europe PMC, ClinicalTrials.gov).
- Evidence-grounded audit pipeline: Europe PMC + web search evidence tiers inform LLM verification to reduce hallucinated rule retention.
- Two-phase data processing pipeline for back-mapping patient diagnoses to recommended diagnostic tests.

Future work

- Continuously improve the audit pipeline to maintain knowledge graph accuracy over time.
- Integration with the SickKids database (where feasible).
- Extend from *back prediction* (diagnosis → test) to *front prediction* — using patient vitals at triage to recommend tests, grounded in the knowledge graph.
- Expert clinician verification of the constructed knowledge graph.